SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	
Register Number	
Class / Slot	Slot B

Capstone Title	
CO1 Evidence Link	
CO2 AT1 VM Recommendation	vCPU: ____ RAM: ____ GB Storage: ____ GB VM Class: ____
CO2 AT2 Evidence Link	
GitHub Repository Link	
Submission Date	

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation		
CO2 AT1 sizing workbook		
VM calibration sheet		
CO2 AT2 practical evidence		
Capstone architecture decision		

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	
Expected workload	
Cloud need	
Virtualization value	
Cost and utilization benefit	
Isolation and flexibility benefit	

Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product.

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

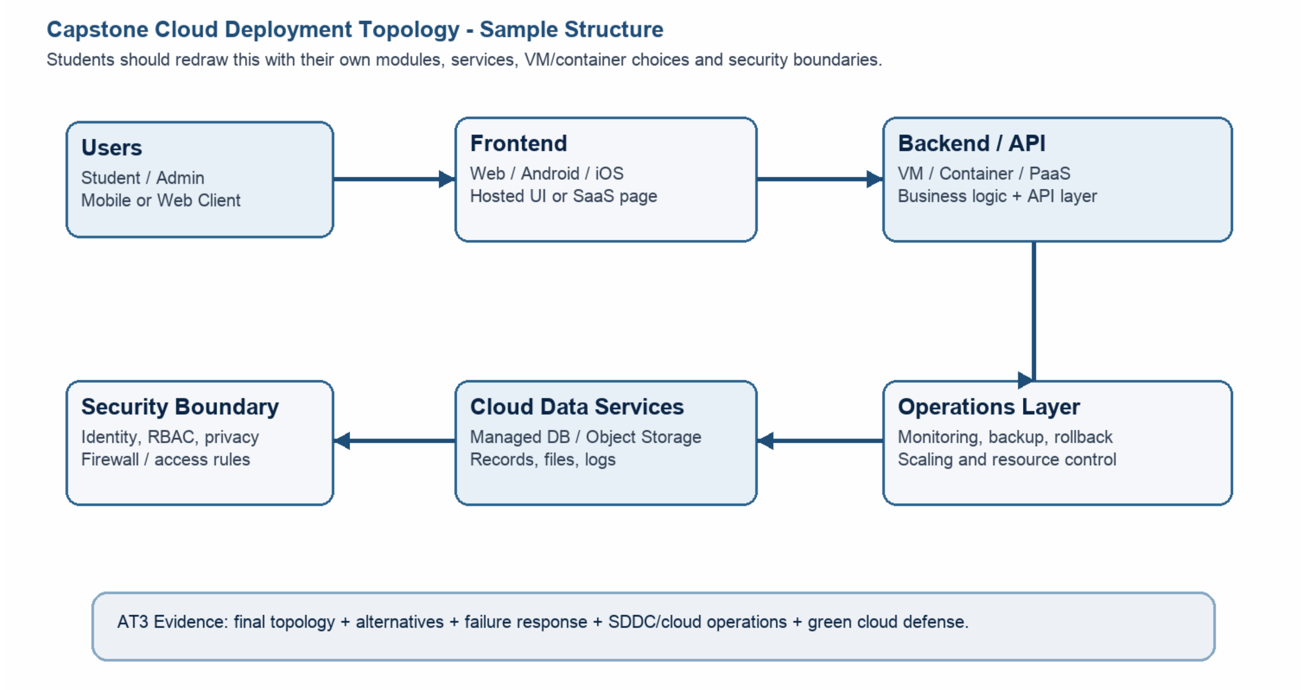
Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment		
Template or repeatable setup		
Snapshot / rollback		
Clone / testing copy		
Container or VM decision		
Maintenance and update plan		

Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk.

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend			
Backend/API			
Database			
Storage			
Authentication			
Notification			
Analytics / AI module			

Paste or draw the final capstone deployment topology diagram here.

SOA/API Integration Explanation: Explain how services communicate and why modular service design is useful.

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1			
Alternative 2			
Selected design			

Security Area	Student Decision
User roles	
Authentication method	
Role-based access control	
Federated access / partner access	
Sensitive data handled	
Privacy protection decision	

Case Analysis Response: Defend why the selected design is better for identity, access and privacy.

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure			
Database growth			
Traffic spike			
Credential leakage			
Service outage			
Resource contention			
Cost increase			

Operational Readiness Area	Student Plan
Automated provisioning	
Software-defined networking	
Software-defined storage	
Monitoring and logging	
Backup and rollback	
Scaling and elasticity	
Resource control and maintenance	
Green Computing and sustainability control	

Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	
VM vs container vs serverless choice	
Scheduled shutdown / idle-resource control	
Log and file retention policy	
Storage lifecycle and backup cleanup	
API-call or polling reduction	
Carbon/energy proxy indicator to track	

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	
Which CO2 concept became clearer after AT1, AT2 and AT3?	
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	
What evidence would you show during viva to defend your infrastructure design?	
What will you improve before final capstone deployment?	

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	
Register Number	
Signature	
Date	

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	